

AI Project 4 Answer Doc
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Question 1 - Definitions

The transition probabilities for rolling and holding:

$$P_{i,j,k,\text{roll}} = 1/6 (1 - P_{j,i,0} + P_{i,j,k+2} + P_{i,j,k+3} + P_{i,j,k+4} + P_{i,j,k+5} + P_{i,j,k+6})$$

$$P_{i,j,k,\text{hold}} = 1 - P_{j,i+k,0}$$

These equations reflect the probabilistic outcomes of either choosing to roll or hold at a given game state, where i is the current player's score, j is the opponent's score, and k is the current turn total. The roll expression accounts for the possibility of losing the turn (rolling a 1), or adding the result to the turn total, while the hold expression models the probability of passing the turn and adding the accumulated turn total to the player's score.

Question 2a - Evaluating $P_{\{99,99,0\}}$

Using the definition from Question 1 (and plugging in 1 for all terminal states) we get:

$$P_{98,99,0} = (1/6)(1 - P_{99,98,0}) + 5/6$$

Before solving algebraically, we will simplify our expression by ruling out one of the terms in the max function.

Let $x = 1 - P_{99,98,0}$. Since $P_{99,98,0} \in [0, 1]$, we know that $x \in [0, 1]$.

Now we compare the two arguments of the max function:

First term: x

Second term: $(1/6)x + 5/6$

We determine which is larger:

$$(1/6)x + 5/6 \geq x$$

$$\Leftrightarrow (1/6)x + 5/6 - x \geq 0$$

$$\Leftrightarrow (-5/6)x + 5/6 \geq 0$$

$$\Leftrightarrow 5/6 \geq (5/6)x$$

$$\Leftrightarrow 1 \geq x$$

This is always true since $x \leq 1$, so:

$$(1/6)(1 - P_{99,98,0}) + 5/6 \geq 1 - P_{99,98,0}$$

with equality only when $P_{99,98,0} = 0$.

Thus, the maximum is always achieved by the second term, and we simplify:

$$P_{98,99,0} = (1/6)(1 - P_{99,98,0}) + 5/6$$

$$P_{99,99,0} = \frac{1}{6}(1 - P_{99,99,0}) + \frac{5}{6}$$

$$\underset{-\frac{5}{6}}{X} = \frac{1}{6}(1 - \underset{-\frac{5}{6}}{X}) + \frac{5}{6}$$

$$\underset{-X}{X - \frac{5}{6}} = \frac{1}{6}(1 - \underset{-X}{X})$$

$$-\frac{5}{6} = \frac{1}{6}(1 - X) - X$$

$$-\frac{5}{6} = \frac{1}{6} - \frac{1}{6}X - X$$

$$-\frac{5}{6} = \frac{1}{6} - \frac{7}{6}X$$

$$\underset{-\frac{1}{6}}{-\frac{5}{6}} \quad \underset{-\frac{1}{6}}{-\frac{1}{6}}$$

$$\underset{-\frac{7}{6}}{-1} = \underset{-\frac{7}{6}}{-\frac{7}{6}X}$$

$$X = \frac{6}{7}$$

$$P_{99,99,0} = \frac{6}{7}$$

Question 2b - Evaluating $P_{\{98,99,0\}}$ and $P_{\{99,98,0\}}$

Using the definition from Question 1 (and plugging in 1 for all terminal states) we get:

$$P_{98,99,0} = (1/6)(1 - P_{99,98,00}) + 5/6$$

$$P_{99,98,0} = (1/6)(1 - P_{98,99,0}) + 5/6$$

Before solving algebraically, we will simplify our expressions by ruling out one of the terms in the max functions (for both expressions).

Let $x = 1 - P_{98,99,0}$. Again, since $P_{98,99,0} \in [0, 1]$, we know $x \in [0, 1]$.
(The same follows if we let $x = 1 - P_{99,98,0}$)

Compare the two arguments:

First term: x

Second term: $(1/6)x + 5/6$

We determine the larger term (Just as we did in part a):

$$(1/6)x + 5/6 \geq x$$

$$\Leftrightarrow (1/6)x + 5/6 - x \geq 0$$

$$\Leftrightarrow (-5/6)x + 5/6 \geq 0$$

$$\Leftrightarrow 5/6 \geq (5/6)x$$

$$\Leftrightarrow 1 \geq x$$

Which is always true. So again, the second term dominates the first (for both).

(except when $P_{98,99,0} = 0$, which cannot happen do to the constant term in our expressions: $5/6$)

Hence:

$$P_{98,99,0} = (1/6)(1 - P_{99,98,00}) + 5/6$$

and

$$P_{99,98,0} = (1/6)(1 - P_{98,99,0}) + 5/6$$

$$p_{98,99,0} = \frac{1}{6} (1 - p_{99,98,0}) + \frac{5}{6}$$

$$p_{99,98,0} = \frac{1}{6} (1 - p_{98,99,0}) + \frac{5}{6}$$

$$X = \frac{1}{6} (1 - y) + \frac{5}{6}$$

$$y = \frac{1}{6} (1 - x) + \frac{5}{6}$$

$$X = \frac{1}{6} \left(1 - \left(\frac{1}{6}X + \frac{5}{6} \right) \right) + \frac{5}{6}$$

$$X = \frac{1}{6} \left(1 - \frac{1}{6}X - \frac{5}{6} \right) + \frac{5}{6}$$

$$X = \frac{1}{6} - \frac{1}{36}X - \frac{5}{36} + \frac{5}{6}$$

$$X = 1 - \frac{5}{36} - \frac{1}{36}X$$

$$\frac{37}{36}X = \frac{31}{36}$$

$$X = \frac{31}{37}$$

$$X = \frac{31}{37}$$

$$y = \frac{1}{6} \left(1 - \frac{6}{37} \right) + \frac{5}{6}$$

$$y = \frac{1}{6} - \frac{6}{222} + \frac{5}{6}$$

$$y = 1 - \frac{1}{37}$$

$$y = \frac{36}{37}$$

$$p_{98,99,0} = \frac{31}{37}$$

$$p_{99,98,0} = \frac{36}{37}$$

Question 2c - Evaluating $P_{\{97,99,0\}}$, $P_{\{97,99,2\}}$, $P_{\{99,97,0\}}$

Using the definition from Question 1 (and plugging in 1 for all terminal states) we get:

1. $P_{97,99,0} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6)$
2. $P_{97,99,2} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + 5/6)$
3. $P_{99,97,0} = \max(1 - P_{97,99,0}, (1/6)(1 - P_{97,99,0}) + 5/6)$

Before solving algebraically, we will simplify each of the expressions by ruling out one of the terms in the max function.

1. $P_{97,99,0} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6)$

We are comparing:

First term: $1 - P_{99,97,0}$

Second term: $(1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6$

Let's assume for contradiction that the first term is the larger one.

Then:

$$1 - P_{99,97,0} > (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6$$

Multiply both sides by 6:

$$6(1 - P_{99,97,0}) > (1 - P_{99,97,0}) + P_{97,99,2} + 4$$

Simplify:

$$5(1 - P_{99,97,0}) > P_{97,99,2} + 4$$

But we know that $P_{97,99,2} \leq 1$, so the right-hand side is at most 5. Then:

$$\begin{aligned} 5(1 - P_{99,97,0}) &> 5 \\ \Rightarrow 1 - P_{99,97,0} &> 1 \\ \Rightarrow P_{99,97,0} &< 0 \end{aligned}$$

This is a contradiction, because $P_{99,97,0} \geq 0$.

So our assumption must be false, and the second term is always greater than or equal to the first. Hence, we simplify:

$$P_{97,99,0} = (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6$$

$$2. \quad P_{97,99,2} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + 5/6)$$

Let $x = 1 - P_{99,97,0}$, with $x \in [0, 1]$.

Compare:

First term: x

Second term: $(1/6)x + 5/6$

As before:

$$(1/6)x + 5/6 \geq x$$

$$\Leftrightarrow (-5/6)x + 5/6 \geq 0$$

$$\Leftrightarrow 1 \geq x$$

So the second term is greater unless $P_{99,97,0} = 0$, in which case both are equal.

Therefore:

$$P_{97,99,2} = (1/6)(1 - P_{99,97,0}) + 5/6$$

$$3. \quad P_{99,97,0} = \max(1 - P_{97,99,0}, (1/6)(1 - P_{97,99,0}) + 5/6)$$

Let $x = 1 - P_{97,99,0}$, with $x \in [0, 1]$.

Compare:

First term: x

Second term: $(1/6)x + 5/6$

Again:

$$(1/6)x + 5/6 \geq x$$

$$\Leftrightarrow (-5/6)x + 5/6 \geq 0$$

$$\Leftrightarrow 1 \geq x$$

So the maximum is achieved by the second term.

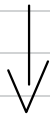
Thus:

$$P_{99,97,0} = (1/6)(1 - P_{97,99,0}) + 5/6$$

$$P_{97,99,0} = \frac{1}{6}(1 - P_{99,97,0}) + \frac{1}{6}P_{97,99,2} + \frac{4}{6}$$

$$P_{97,99,2} = \frac{1}{6}(1 - P_{99,97,0}) + \frac{5}{6}$$

$$P_{99,97,0} = \frac{1}{6}(1 - P_{97,99,0}) + \frac{5}{6}$$



$$X = \frac{1}{6}(1 - Z) + \frac{1}{6}Y + \frac{4}{6}$$

$$Y = \frac{1}{6}(1 - Z) + \frac{5}{6}$$

$$Z = \frac{1}{6}(1 - X) + \frac{5}{6}$$

$$X = \frac{1}{6}(1-z) + \frac{1}{6}y + \frac{4}{6}$$

$$X = \frac{1}{6} \left(1 - \left(\frac{1}{6}(1-x) + \frac{5}{6} \right) \right) + \frac{1}{6} \left(\frac{1}{6} \left(1 - \left(\frac{1}{6}(1-x) + \frac{5}{6} \right) \right) + \frac{5}{6} \right) + \frac{4}{6}$$

$$X = \frac{1}{6} \left(1 - \frac{1}{6} + \frac{1}{6}x - \frac{5}{6} \right) + \frac{1}{6} \left(\frac{1}{6} \left(1 - \frac{1}{6} + \frac{1}{6}x - \frac{5}{6} \right) + \frac{5}{6} \right) + \frac{4}{6}$$

$$X = \frac{1}{6} \left(\frac{1}{6}x \right) + \frac{1}{6} \left(\frac{1}{6} \left(\frac{1}{6}x \right) + \frac{5}{6} \right) + \frac{4}{6}$$

$$X = \frac{1}{6} \left(\frac{1}{6}x \right) + \frac{1}{6} \left(\frac{1}{36}x + \frac{5}{6} \right) + \frac{4}{6}$$

$$X = \frac{1}{36}x + \frac{1}{216}x + \frac{5}{36} + \frac{4}{6}$$

$$X = \frac{7}{216}x + \frac{29}{36}$$

$$+ \frac{7}{216}x + \frac{7}{216}x$$

$$\frac{223}{216}x = \frac{29}{36}$$

$$\frac{223}{216} \quad \frac{223}{216}$$

$$X = \frac{6264}{8028}$$

$$X = \frac{174}{223}$$

$$Z = \frac{1}{6}(1-x) + \frac{5}{6}$$

$$Z = \frac{1}{6} \left(1 - \frac{174}{223} \right) + \frac{5}{6}$$

$$Z = \frac{1}{6} \left(\frac{49}{223} \right) + \frac{5}{6}$$

$$Z = \frac{49}{1338} + \frac{5}{6}$$

$$Z = \frac{49}{1338} + \frac{1115}{1338}$$

$$Z = \frac{1164}{1338}$$

$$Z = \frac{194}{223}$$

$$y = \frac{1}{6}(1-z) + \frac{5}{6}$$

$$y = \frac{1}{6} \left(1 - \frac{194}{223} \right) + \frac{5}{6}$$

$$y = \frac{1}{6} \left(\frac{29}{223} \right) + \frac{5}{6}$$

$$y = \frac{29}{1338} + \frac{5}{6}$$

$$y = \frac{29}{1338} + \frac{1115}{1338}$$

$$y = \frac{1144}{1338}$$

$$y = \frac{572}{669}$$

$$P_{97, 99, 0} = \frac{174}{223}$$

$$P_{97, 99, 2} = \frac{572}{669}$$

$$P_{99, 97, 0} = \frac{194}{223}$$

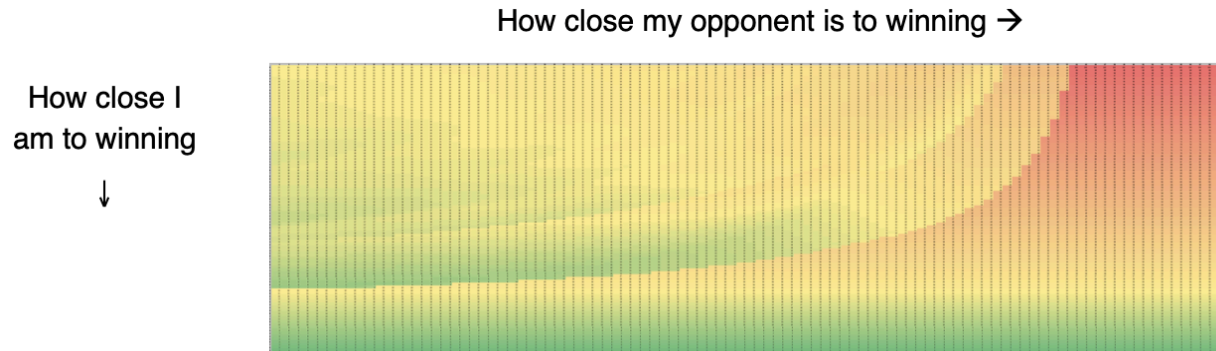
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Question 4:

- a) The probability that the first player wins assuming both players play optimally from the start is calculated by our algorithm. When our current score, $i = 0$ and the opponent's score, $j = 0$, and the current turn, $k = 0$, then the probability of the first player winning is 0.5305927250129694. We define optimality here by assuming that all of a player's rolls will be in favor of them winning starting from the first turn. Essentially, when the game has just started and both the player and their opponent's score is 0, and player 1 gets the best rolls, the chance to win is 0.5305927250129694.
- b) On the first turn, the lowest turn total where the player should hold rather than roll is 21. This is determined by the algorithm which calculates the probabilities of winning and chooses the max.
- c) If player 1 scores no points on their first turn the probability that the second player wins is the same as with 4a, 0.5305927250129694. This is because the same conditions apply where player 1 now has no points and player 2 plays optimally from their turn 1. For the same reason, the hold threshold for player 2 matches 4b, as it is also 21.

Question 5 - Heatmap of results



Description of heatmap:

- When a player is far behind and the opponent is close to winning (upper right), the map is yellow to red, indicating a **high hold threshold**.
 - This means it makes more sense to **take greater risks**, since being conservative will likely lead to losing.
- As a player's score improves and they get closer to winning (toward the bottom), the map becomes green, suggesting a **lower hold threshold**.
 - This shows that it is better to **play cautiously** and protect your lead.
- There is a **transition zone** where the color shifts again, signaling that once a player has gained a strong enough lead, they **can afford to take more risks**.